llmXive Automated Review of MMSkills: Towards Multimodal Skills for General Visual Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (13 lenses): paper_reviewer, Claim Accuracy, Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, Writing Quality, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

paper_reviewer

Focuses narrowly on the aspect of the manuscript.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer.md`

Free-form review body

Strengths

- **Novel Representation**: The concept of "multimodal procedural knowledge" (binding text, state cards, and visual evidence) is a significant step forward from text-only skill libraries. The formalization of state cards with "when-to-use" and "verification cues" is well-motivated.
- **Branch-Loading Mechanism**: The two-stage branch-loading approach (gated view selection + planning) is a clever engineering solution to the "context pollution" and "visual anchoring" problems inherent in loading large multimodal skill packages. The ablation studies in Figure 3 effectively demonstrate the necessity of both components.
- **Comprehensive Evaluation**: The paper evaluates across diverse benchmarks (OSWorld, macOSWorld, VAB-Minecraft, Super Mario) and model sizes (from 8B to 235B parameters), providing a robust picture of the method's generalizability.
- **Behavioral Analysis**: The analysis of agent behavior (RQ4) goes beyond success rates, showing that MMSkills reduce repetitive actions and low-level primitive usage, which is a strong qualitative indicator of improved reasoning.

Concerns

- **Critical Data Leakage Risk (Major)**: The most significant flaw is the potential overlap between the evaluation benchmarks (OSWorld, macOSWorld) and the skill source (OpenCUA).
- OSWorld and macOSWorld are standard benchmarks for computer-use agents.
- OpenCUA is described as a source of "public non-evaluation trajectories" for Ubuntu and macOS.
- Both datasets cover the same application domains (Chrome, LibreOffice, VS Code, etc.).
- The paper claims in Appendix A.2 that "All MMSkills are extracted from non-test trajectories" and that source data is "disjoint from the final evaluation cases." However, without a rigorous deduplication check (e.g., comparing task instructions, screenshots, or trajectory hashes), it is highly probable that the "source" trajectories contain the exact same tasks or very similar variations as the "test" set. If the agent is effectively "memorizing" the test tasks from the source data, the reported improvements are invalid. The current evidence (just stating they are disjoint) is insufficient for a scientific claim of generalization.
- **Bibliography Integrity**: The bibliography entry for the paper itself (arXiv 2605.13527) is marked as "unreachable" in the provided metadata. While this might be a metadata artifact, the reference list contains several 2025/2026 citations (e.g., `wang2025opencua`, `yang2025macosworld`). The reviewer must verify that these are real, accessible preprints or published works. If they are future-dated or inaccessible, the scientific grounding is weak.

- **Skill Generation Complexity**: The "Generator" pipeline (5 phases) is complex. While the paper describes it, the reproducibility of the skill generation process is unclear. How are "meta-skills" defined? How is the "audit" phase automated? The lack of a detailed algorithm or pseudocode for the Generator (beyond the high-level pipeline) makes it hard to assess the quality of the generated skills.
- **Cost Analysis**: The paper mentions "extra inference cost from branch loading" as a limitation but does not quantify it. Given the two-stage branch process and the loading of images, the latency and token cost increase could be significant. A brief analysis of the trade-off between performance gain and computational cost would strengthen the paper.

Recommendation

The paper presents a compelling and well-structured framework for multimodal skills, with strong experimental results and a novel branch-loading mechanism. However, the **validity of the experimental results is currently in question due to the high risk of data leakage** between the skill source (OpenCUA) and the evaluation benchmarks (OSWorld/macOSWorld). The claim that the source and test sets are "disjoint" is not sufficiently supported by evidence.

Verdict: major_revision_science

The authors must:

1. Perform a rigorous deduplication analysis between the source trajectories and the test sets. If any overlap is found, the experiments must be re-run with a strictly filtered source set.
2. Provide a detailed explanation of how the "disjoint" property was ensured (e.g., hash matching, semantic similarity thresholds).
3. Clarify the provenance and accessibility of the cited 2025/2026 works.
4. Add a section or table quantifying the inference cost/latency overhead of the branch-loading mechanism.

Once these scientific concerns are addressed, the paper will be a strong contribution to the field of visual agents.

Action items.

- [**science**] Branch-Loading Mechanism: The two-stage branch-loading approach (gated view selection + planning) is a clever engineering solution to the "context pollution" and "visual anchoring" problems inherent in loading large multimodal skill packages. The ablation studies in Figure 3 effectively demonstrate the necessity of both components.
- [**science**] Comprehensive Evaluation: The paper evaluates across diverse benchmarks (OS-World, macOSWorld, VAB-Minecraft, Super Mario) and model sizes (from 8B to 235B parameters), providing a robust picture of the method's generalizability.
- [**science**] Behavioral Analysis: The analysis of agent behavior (RQ4) goes beyond success rates, showing that MMSkills reduce repetitive actions and low-level primitive usage, which is a strong qualitative indicator of improved reasoning. ## Concerns

- **[science]** Critical Data Leakage Risk (Major): The most significant flaw is the potential overlap between the evaluation benchmarks (OSWorld, macOSWorld) and the skill source (OpenCUA).
- **[science]** OSWorld and macOSWorld are standard benchmarks for computer-use agents.
- **[science]** OpenCUA is described as a source of "public non-evaluation trajectories" for Ubuntu and macOS.
- **[science]** Both datasets cover the same application domains (Chrome, LibreOffice, VS Code, etc.).
- **[science]** The paper claims in Appendix A.2 that "All MMSkills are extracted from non-test trajectories" and that source data is "disjoint from the final evaluation cases." However, without a rigorous deduplication check (e.g., comparing task instructions, screenshots, or trajectory hashes), it is highly probable that the "source" trajectories contain the exact same tasks or very similar variations as the "test" set. If the agent is effectively "memorizing" the test tasks from the source data, the reported
- **[science]** Bibliography Integrity: The bibliography entry for the paper itself (arXiv 2605.13527) is marked as "unreachable" in the provided metadata. While this might be a metadata artifact, the reference list contains several 2025/2026 citations (e.g., wang2025opencua, yang2025macosworld). The reviewer must verify that these are real, accessible preprints or published works. If they are future-dated or inaccessible, the scientific grounding is weak.
- **[science]** Skill Generation Complexity: The "Generator" pipeline (5 phases) is complex. While the paper describes it, the reproducibility of the skill generation process is unclear. How are "meta-skills" defined? How is the "audit" phase automated? The lack of a detailed algorithm or pseudocode for the Generator (beyond the high-level pipeline) makes it hard to assess the quality of the generated skills.
- **[science]** Perform a rigorous deduplication analysis between the source trajectories and the test sets. If any overlap is found, the experiments must be re-run with a strictly filtered source set.
- **[science]** Provide a detailed explanation of how the "disjoint" property was ensured (e.g., hash matching, semantic similarity thresholds).
- **[science]** Clarify the provenance and accessibility of the cited 2025/2026 works.
- **[science]** Add a section or table quantifying the inference cost/latency overhead of the branch-loading mechanism. Once these scientific concerns are addressed, the paper will be a strong contribution to the field of visual agents.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the consistency between reported statistics, tables, and textual assertions.

Statistical Consistency in Results Tables: There are potential discrepancies between the domain-level success rates and the reported "Overall" success rates in Table 1 (OSWorld) and Table 2 (macOSWorld).

1. **Table 1 (OSWorld):** For Qwen3-VL-235B under MMSkills, the reported overall success is 39.17%. A weighted calculation using the domain counts from Appendix Table A1 (e.g., Chrome: 45, GIMP: 26, etc.) and the domain success rates in Table 1 does not immediately yield 39.17%. For instance, the high performance in Chrome (59.91%) and GIMP (69.23%) should pull the average up significantly. The current overall figure seems lower than a simple weighted average of the visible domain scores. The authors must verify if the "Overall" column is a weighted average, a simple mean, or derived from a different set of raw counts not shown in the table.
2. **Table 2 (macOSWorld):** Similar concerns exist for the macOSWorld "Overall" column. The text claims MMSkills improve performance, but the specific aggregate numbers need to be mathematically consistent with the domain breakdowns provided in Appendix Table A1.

Claim Verification in Text vs. Data:

1. **Step Reduction (Section 4.3):** The text claims the "largest reductions" in interaction steps appear for Qwen3-VL-235B. While Table 3 shows a 5.35 step reduction on OSWorld for this model, the same table shows a 7.67 step reduction on VAB-Minecraft for the same model. The text phrasing "largest reductions appearing for Qwen3-VL-235B" is ambiguous; it should specify "on OSWorld" or acknowledge the even larger reduction on VAB-Minecraft to avoid misleading the reader about the magnitude of the effect across all benchmarks.
2. **Behavioral Shift Metrics (Section 4.4):** The claim that "exact repeated actions fall from 21.8% to 6.2%" for Qwen3-VL-235B is specific. However, Figure 3 (Panel C) does not explicitly label these percentages on the axis or in the caption. While the visual trend supports the claim, the specific numbers should be explicitly cited in the figure caption or a supplementary table to ensure the claim is fully supported by the visual evidence presented.

Citation Accuracy: The citations for benchmarks (OSWorld, macOSWorld, VAB-Minecraft, LMGame-Bench) and related work (SkillWeaver, Voyager, etc.) appear to match the standard literature. However, several citations (e.g., `yang2025macosworld`, `hu2025lmgamebench`, `bai2025qwen3vl`) refer to 2025/2026 dates. Given the current date context, these are likely preprints or future-dated arXiv submissions. The reviewer cannot verify the existence of these specific versions without access to the live arXiv, but the formatting suggests they are treated as established works. If these are indeed very recent or future-dated, the authors should ensure the citation keys match the actual available versions to avoid "unreachable" or "future" citation errors in the final PDF.

Recommendation: The authors should re-calculate the "Overall" success rates in Tables 1 and 2 to ensure they match the weighted averages of the domain-level data. Additionally, clarify the "largest reduction" claim in Section 4.3 to account for the VAB-Minecraft results, and explicitly label the specific percentage values in Figure 3 or its caption.

Action items.

- **[writing]** Table 1 (OSWorld): For Qwen3-VL-235B under MMSkills, the reported overall success is 39.17%. A weighted calculation using the domain counts from Appendix Table A1 (e.g., Chrome: 45, GIMP: 26, etc.) and the domain success rates in Table 1 does not immediately yield 39.17%. For instance, the high performance in Chrome (59.91%) and GIMP (69.23%) should pull the average up significantly. The current overall figure seems lower than a simple weighted average of the visible domain scores. The authors m
- **[writing]** Table 2 (macOSWorld): Similar concerns exist for the macOSWorld "Overall" column. The text claims MMSkills improve performance, but the specific aggregate numbers need to be mathematically consistent with the domain breakdowns provided in Appendix Table A1. Claim Verification in Text vs. Data:
- **[writing]** Step Reduction (Section 4.3): The text claims the "largest reductions" in interaction steps appear for Qwen3-VL-235B. While Table 3 shows a 5.35 step reduction on OSWorld for this model, the same table shows a 7.67 step reduction on VAB-Minecraft for the same model. The text phrasing "largest reductions appearing for Qwen3-VL-235B" is ambiguous; it should specify "on OSWorld" or acknowledge the even larger reduction on VAB-Minecraft to avoid misleading the reader about the magnitude of the effec
- **[writing]** Behavioral Shift Metrics (Section 4.4): The claim that "exact repeated actions fall from 21.8% to 6.2%" for Qwen3-VL-235B is specific. However, Figure 3 (Panel C) does not explicitly label these percentages on the axis or in the caption. While the visual trend supports the claim, the specific numbers should be explicitly cited in the figure caption or a supplementary table to ensure the claim is fully supported by the visual evidence presented. Citation Accuracy: The citations for benchmarks (OS

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

This re-review confirms that the four specific claim-accuracy issues identified in the prior review remain unaddressed in the current revision. The manuscript continues to present aggregate statistics in Tables 1 and 2 that do not mathematically reconcile with their constituent domain breakdowns, creating a risk of reader confusion regarding the true performance gains. Additionally, the text in Section 4.3 retains an ambiguous claim about "largest reductions" that contradicts the data in Table 3, and Section 4.4 still cites specific behavioral shift percentages (21.8% to

6.2%) that are not explicitly visible in the referenced Figure 3. These are not stylistic issues but factual inconsistencies where the stated evidence does not support the specific numbers or superlatives used. The authors must perform the necessary calculations to align the "Overall" figures with the domain data and clarify the textual claims to match the tabular evidence.

Action items.

- [**writing**] This re-review confirms that the four specific claim-accuracy issues identified in the prior review remain unaddressed in the current revision. The manuscript continues to present aggregate statistics in Tables 1 and 2 that do not mathematically reconcile with their constituent domain breakdowns, creating a risk of reader confusion regarding the true performance gains. Additionally, the text in Section 4.3 retains an ambiguous claim about "largest reductions" that contradicts the data in Table 3

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively visualizes the prompt surfaces and workflow of the branch-loaded multimodal skill agent. The diagram is clear, with distinct sections for the Main Visual Agent, Branch Stage 1, and Branch Stage 2, and the flow of information is well-represented by arrows and color coding. The caption accurately describes the figure's content.

Figure 2

The figure is visually clear and the panels are well-organized, but the caption contains a broken cross-reference to a missing figure number, and the x-axis label in Panel B is slightly ambiguous regarding the specific metric definition.

Figure 3

The figure clearly visualizes the interaction turns for the LibreOffice Calc task, but the caption's claim about color-coded labels is not supported by a visible legend or key within the figure itself.

Figure 4

The figure provides a clear, step-by-step visualization of the terminal task workflow. However, the caption's claim that colored labels distinguish specific categories is not supported by a visible legend or key within the figure itself.

Figure 5

The figure presents ablation data clearly, but the x-axis labels in Panel (B) are ambiguous regarding the specific ablation conditions, and the legend symbols for component presence are not explicitly defined.

Figure 6

Figure 6 effectively visualizes behavioral shifts, but Panel (B) contains a mismatch between the legend entries and the plotted data rows, and there is an inconsistency in model naming between the legend and other panels.

Figure 7

Figure 7 effectively illustrates the MMSkills framework by contrasting a multimodal skill package (Panel A) with three agent decision paths (Panel B). The visual layout clearly demonstrates how branch-loaded multimodal skills succeed where text-only or no-skill approaches fail, aligning perfectly with the provided caption.

Figure 8

Figure 8 provides a clear and comprehensive overview of the MMSkills framework, effectively illustrating the data flow from the skill package and generation pipeline to the branch-loaded agent architecture. The visual hierarchy and labeling are consistent with the provided caption, making the complex system easy to understand.

Action items.

- [**writing**] Figure 2 caption contains a broken cross-reference: 'The panels follow the same metrics as Figure :' is missing the target figure number (likely Figure 6).
- [**science**] Figure 2 Panel (B) x-axis label 'Mean count per task' is ambiguous; the caption describes it as 'low-level primitives per task', but the axis lacks the specific unit or definition of 'count'.
- [**writing**] Figure 3: The caption claims 'Colored turn labels distinguish direct GUI actions, skill loading, branch guidance, evidence-gated reasoning, and final completion,' but the figure lacks a legend or key to map the specific colors (orange, green, blue, purple) to these categories.
- [**writing**] Figure 4: The caption claims 'Colored turn labels distinguish direct GUI actions, skill loading, branch guidance, evidence-gated reasoning, and final completion,' but the figure lacks a legend or key mapping the specific colors (e.g., orange, green, purple) to these categories.
- [**writing**] Figure 4: The top section lists 'GUI action' and 'Branch guidance' as categories,

but the 'Branch guidance' label is not used in the turn sequence (Turns 2B and 5B are labeled 'Branch'), creating a minor inconsistency between the legend and the timeline.

- **[science]** Figure 5: The x-axis labels in Panel (B) ('Direct-Full', 'Direct-Selected', 'Branch-Full') are ambiguous and do not clearly map to the specific ablation conditions described in the caption (e.g., 'branch loading with or without view selection'), making it difficult to interpret the specific contribution of each component.
- **[writing]** Figure 5: The checkmarks and 'x' symbols in the legend table are not explicitly defined in the caption or figure text, requiring the reader to infer that they represent the presence or absence of 'Cards' and 'Images'.
- **[science]** Figure 6, Panel (B): The legend lists 'Qwen3-VL-235B / No skill' and 'Qwen3-VL-235B / MMSkills', but the y-axis labels only include 'All primitives / task', 'Clicks / task', 'Keyboard / task', and 'Scroll+wait / task'. The 'All primitives' row has no corresponding data points for the Qwen models, creating a mismatch between the legend and the plotted data.
- **[writing]** Figure 6, Panel (B): The legend uses 'Qwen3-VL-235B' while the main text and other panels (A, C) refer to 'Qwen'. This inconsistency in model naming (Qwen vs Qwen3-VL-235B) across the figure panels is confusing and should be standardized.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper introduces a novel framework, MMSkills, but relies heavily on a dense system of mathematical notation and pipeline shorthand that is not defined at first use. While the concepts are sound, a competent reader from an adjacent field (e.g., NLP or robotics) will stall repeatedly when encountering the equations in Sections 2.1 through 2.4.

Specifically, Equation 1 introduces O_t, H_t, $\mathcal{C}_I$, and A_t without any accompanying text defining them as observation, history, candidate skills, and action. Similarly, the generator pipeline in Equation 5 uses a string of set symbols ($\mathcal{T}_d, \mathcal{C}_d, \dots$) that are never explicitly mapped to their semantic content (e.g., "clustered trajectories"). The view selection variables J_t and R_t in Equation 6 are also undefined.

These are not standard field-wide terms but specific notation for this paper's method. The fix is straightforward: add a single sentence or clause immediately following each equation to define the symbols used. This will make the method section self-contained and accessible to the target audience without altering the scientific content.

Action items.

- **[writing]** Section 2.1 (Eq. 1) introduces symbols O_t, H_t, $\mathcal{C}_I$, and A_t without definition. An adjacent-field reader cannot infer that O_t is the current observation, H_t is the history, or $\mathcal{C}_I$ is the candidate skill set. Add a clause immediately following Eq. 1 defining each symbol (e.g., 'where O_t is the current screenshot, H_t is the interaction history...').

- [**writing**] Section 2.2 (Eq. 3) uses the symbol $\mathcal{V}_j$ and the term 'available_views' without defining the set of valid view types. While 'full_frame' etc. appear in Eq. 4, the reader encounters $\mathcal{V}_j$ first without knowing it represents a set of view identifiers. Define $\mathcal{V}_j$ explicitly in the text preceding Eq. 3.
- [**writing**] Section 2.3 (Eq. 5) introduces $\mathcal{T}_d$, $\mathcal{C}_d$, $\mathcal{A}_d$, $\mathcal{R}_d$, and $\widehat{\mathcal{M}}_d$ as intermediate sets in the generator pipeline without defining what each set contains (e.g., clustered trajectories, abstracted procedures). Add a sentence defining these intermediate variables before or after Eq. 5.
- [**writing**] Section 2.4 (Eq. 6) uses J_t and R_t without definition. The text says 'Stage 1 (view selection)' but does not explicitly state that J_t is the set of selected skill indices and R_t is the set of selected view types. Define these variables in the text immediately preceding Eq. 6.
- [**writing**] Section 2.3 introduces 'Phase 0' through 'Phase 4' in Eq. 5 and the surrounding text, but does not define what each phase does (e.g., 'Phase 0: clustering', 'Phase 1: abstraction'). An adjacent reader cannot follow the pipeline flow. Add a brief parenthetical or list defining the function of each phase.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent high-level argument: visual agents require multimodal procedural knowledge (state + text + images) rather than text-only skills, and a branch-loading mechanism is necessary to avoid context overload. However, several internal logical gaps exist between the stated mechanisms and the supporting evidence or definitions.

First, the **Skill Generation Pipeline** (Section 3.2, Eq. 4) claims to "merge overlapping candidates" in Phase 2 to produce generalized skills. The logical consistency of this claim is undermined by the absence of a defined metric for "overlap" in a multimodal context. Does overlap rely on textual procedure similarity, visual state similarity, or a combination? Without specifying the logical criteria for merging, the claim that the pipeline effectively generalizes skills rather than simply aggregating them remains an unsupported assertion.

Second, there is a **tension in the results interpretation** regarding efficiency. Section 4.2 argues that MMSkills shorten trajectories by helping agents avoid "unnecessary exploration." However, Table 3 shows that MMSkills significantly increase the *frequency* of skill invocation (e.g., Qwen3-235B on OSWorld: calls/case rises from 0.49 to 0.92). The paper fails to logically bridge this gap: it does not explicitly demonstrate that the *cost* of the additional consultations is outweighed by the *savings* in failed actions. The conclusion that "more calls = fewer steps" requires a specific argument about the efficiency of each consultation, which is currently missing.

Third, the **mechanism for preventing "over-anchoring"** (Section 3.3) is asserted but not logically validated. The paper claims branch loading prevents the agent from planning around reference screenshots. Yet, the Stage 2 prompt (Appendix A.3) explicitly instructs the model to "align the selected evidence with the live state." The paper does not provide a logical argument or ablation study showing that this alignment instruction successfully mitigates anchoring compared to a control where the model is simply told to ignore the images. The causal link between the "branch" architecture and the reduction of anchoring bias is assumed rather than derived from the presented evidence.

These issues do not invalidate the core contribution but require clarifying the logical steps connecting the proposed mechanisms to the observed outcomes.

Action items.

- [**science**] In Section 3.2 (Eq. 4), the generator pipeline claims to 'merge overlapping candidates' in Phase 2, yet the text does not specify the logical criteria for determining overlap between multimodal packages (e.g., visual similarity vs. textual procedure). Without this definition, the claim that the pipeline produces 'generalized' skills lacks a supporting mechanism.
- [**writing**] Section 4.2 claims MMSkills reduce interaction steps by avoiding 'unnecessary exploration,' but Table 3 shows that MMSkills increase the average number of skill calls per case (e.g., Qwen3-235B: 0.49 -> 0.92). The paper does not logically reconcile how increased consultation frequency leads to reduced total steps without explicitly arguing that each consultation prevents multiple failed actions.
- [**science**] The 'Branch Loading' mechanism (Section 3.3) asserts that it prevents 'over-anchoring' to reference screenshots. However, the prompt templates in Appendix A.3 (Stage 2) instruct the model to 'align the selected evidence with the live state.' The paper lacks a logical argument or ablation proving that this alignment process successfully prevents anchoring compared to a baseline where the model simply ignores the reference images.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

This re-review confirms that the three logical consistency issues identified in the prior review remain unaddressed in the current manuscript. The authors have not provided the missing definitions, reconciliations, or ablation arguments required to close the gaps in their reasoning.

1. **Undefined Overlap Criteria (Item 5221a0349f87):** In Section 3.2, the generator pipeline describes a "merge overlapping candidates" step in Phase 2. However, the manuscript does

not specify the logical or mathematical criteria used to determine if two multimodal packages "overlap." Is it based on visual similarity of keyframes, textual similarity of procedures, or semantic overlap of state cards? Without this definition, the claim that the pipeline produces "generalized" skills is unsupported, as the mechanism for generalization (merging) is undefined.

2. **Unreconciled Metric Contradiction (Item a813b195680b):** Section 4.2 argues that MMSkills reduce total interaction steps by avoiding "unnecessary exploration." However, Table 3 explicitly shows that the "Calls/case" metric increases from 0.71 (Text-only) to 0.96 (MMSkills) for the Gemini 3 Flash model. The text does not provide the necessary logical bridge to explain how *more* skill consultations result in *fewer* total steps. The argument implicitly assumes that each consultation prevents multiple failed actions, but this assumption is not stated or supported by the data presented.

3. **Unproven Anti-Anchoring Mechanism (Item f41df7885fd2):** Section 3.3 claims the "Branch Loading" mechanism prevents "over-anchoring" to reference screenshots. Conversely, the prompt templates in Appendix A.3 (Stage 2) explicitly instruct the model to "align the selected evidence with the live state." The paper offers no logical argument or ablation study demonstrating that this alignment process successfully prevents anchoring compared to a baseline where the model ignores reference images. The claim of preventing a specific failure mode (anchoring) is not entailed by the described mechanism (alignment).

These issues represent breaks in the chain of reasoning where conclusions do not strictly follow from the premises or evidence provided. Addressing them requires either adding the missing definitions/arguments (science) or clarifying the text to match the data (writing).
Action items.

- [**writing**] Undefined Overlap Criteria (Item 5221a0349f87): In Section 3.2, the generator pipeline describes a "merge overlapping candidates" step in Phase 2. However, the manuscript does not specify the logical or mathematical criteria used to determine if two multimodal packages "overlap." Is it based on visual similarity of keyframes, textual similarity of procedures, or semantic overlap of state cards? Without this definition, the claim that the pipeline produces "generalized" skills is unsupported, as

- [**writing**] Unreconciled Metric Contradiction (Item a813b195680b): Section 4.2 argues that MMSkills reduce total interaction steps by avoiding "unnecessary exploration." However, Table 3 explicitly shows that the "Calls/case" metric increases from 0.71 (Text-only) to 0.96 (MMSkills) for the Gemini 3 Flash model. The text does not provide the necessary logical bridge to explain how *more* skill consultations result in *fewer* total steps. The argument implicitly assumes that each consultation prevents multip

- [**writing**] Unproven Anti-Anchoring Mechanism (Item f41df7885fd2): Section 3.3 claims the "Branch Loading" mechanism prevents "over-anchoring" to reference screenshots. Conversely, the prompt templates in Appendix A.3 (Stage 2) explicitly instruct the model to "align the selected evidence with the live state." The paper offers no logical argument or ablation study demonstrating that this alignment process successfully prevents anchoring compared to a baseline where the model ignores reference images. The cl

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a compelling framework for multimodal skills, but the rhetoric in the abstract and conclusion occasionally exceeds the specific scope of the reported experiments.

First, the abstract asserts "consistent improvements across model families." While the text mentions testing GLM-5V and Kimi-K2.6 (Appendix Fig 2), the primary results table (Table 1) only explicitly displays data for Gemini 3.1 Pro and Qwen3-VL-235B. The claim of "consistency" is not fully licensed by the highlighted data in the main body, as the reader cannot verify the trend for the other families from the primary evidence presented. The authors should either include the full comparative data in the main table or hedge the claim to "tested model families" to avoid implying a universal trend that isn't visually summarized in the key results.

Second, the scope of "general visual agents" in the title and abstract is broader than the evidence. The experiments are confined to desktop GUI environments (OSWorld, macOSWorld) and two specific game environments (Minecraft, Mario). There is no evidence presented for embodied agents (robotics), video analysis, or other visual domains. While the method *could* apply there, the paper's rhetoric suggests a broader validation than the "screen-based and game" reality of the data. A more precise limitation acknowledging that the "multimodal procedural knowledge" was only validated in screen-based contexts would align the claim with the evidence.

Finally, the conclusion states the method "improves GUI and game agents," which is accurate, but the abstract's framing of "complementing internal priors" is a strong theoretical claim. The paper demonstrates this complementarity only within the specific constraints of the OpenCUA-derived skills and the tested benchmarks. The claim is not false, but it is presented as a general principle rather than a finding specific to the tested domain. Narrowing the language to reflect the specific domain of validation (screen-based tasks) would tighten the link between the rhetoric and the demonstrated scope.

These are primarily issues of framing and precision. The paper is a solid contribution, but tightening the scope of the claims to match the specific experimental boundaries would eliminate the overreach.

Action items.

- [**writing**] Abstract claims 'consistent improvements across model families,' but Table 1 only shows Gemini and Qwen. Include results for GLM/Kimi in the main table or hedge to 'tested families' to support the generalization.
- [**writing**] Title/Abstract imply 'general visual agents,' but experiments are limited to GUI and two games. Add a limitation explicitly restricting the scope to screen-based and game

environments, excluding robotics or video.

- [**writing**] Abstract claims the method 'complements internal priors' generally, but evidence is limited to OpenCUA-derived skills on specific benchmarks. Narrow the claim to 'complements priors in screen-based task execution'.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a framework for multimodal skills in visual agents, utilizing screenshots and state cards derived from public trajectories. From a safety and ethics perspective, the work is low-risk. The authors explicitly address the primary safety concern in the "Broader Impact" section (Appendix), acknowledging that storing screenshots carries privacy risks and stating that skills are constructed from "public non-evaluation trajectories" to avoid private data.

The methodology relies on existing public benchmarks (OSWorld, macOSWorld) and public trajectory datasets (OpenCUA), which are standard in the field and do not involve new human-subjects data collection requiring IRB approval. The "Use of LLMs" section clarifies the role of models in generation and evaluation without raising undisclosed conflicts.

While the capability to automate desktop tasks is dual-use, the paper does not provide operational details for exploiting specific vulnerabilities, nor does it claim to bypass security controls. The "branch-loaded" mechanism is a retrieval-augmented generation technique, not a tool for deception or surveillance. The authors' brief discussion of privacy and the reliance on public data sources constitute an adequate disclosure for this type of research. No specific, non-trivial risks were identified that require further mitigation or disclosure.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling framework for multimodal skills, but the experimental design currently lacks the rigor required to support the headline claims of consistent improvement.

1. Lack of Variance Reporting (Single-Run Risk): The primary results in Table 1 (OSWorld) and Table 2 (macOSWorld) report single-point accuracy percentages (e.g., 50.11% vs 44.08%). There is no mention of the number of random seeds used, standard deviations, or confidence intervals. In visual agent benchmarks, performance can fluctuate significantly based on the specific test cases encountered or the stochasticity of the LLM's generation. A 6-point gain on a 360-example test set is statistically fragile without variance estimates. The authors must report results averaged over at least 3-5 seeds to demonstrate that the improvement is a stable effect and not a lucky draw of the test set or a single favorable run.

2. Confounded Baselines (Mechanism vs. Content): The comparison between "No skill" and "Text-only" in Table 1 reveals a critical design flaw. The "Text-only" condition (40.76%) actually underperforms the "No skill" baseline (44.08%). This suggests that the *mechanism* of loading skills (the branch-loading architecture) or the specific prompt templates used introduce overhead or confusion that degrades performance. The paper attributes the final gain (50.11%) entirely to the "multimodal" nature of the skills, but it fails to isolate the contribution of the *content* (images/cards) from the *cost* of the *mechanism*. Without a "Branch-only" control (using the branch mechanism but with no skill content or a neutral placeholder), the reader cannot determine if the multimodal skills are actually helpful or if the system is just less broken than the text-only version.

3. Ablation Validity: Figure 2 presents ablations removing state cards or keyframes. However, the text does not clarify if these ablations control for the *amount* of context or the *complexity* of the prompt. If the "Text-only" ablation simply removes images but keeps the same token budget and branch structure, it is a fair test. If, however, the "Text-only" baseline in the main table is a naive concatenation while the ablation uses the branch loader, the comparison is invalid. The authors need to explicitly state that the ablation conditions differ *only* in the presence of visual evidence, holding the branch-loading mechanism and prompt structure constant.

To accept these claims, the authors must provide multi-seed variance data and disentangle the performance cost of the branch-loading mechanism from the benefit of the multimodal content.

Action items.

- [**science**] The paper presents a compelling framework for multimodal skills, but the experimental design currently lacks the rigor required to support the headline claims of consistent improvement. 1. Lack of Variance Reporting (Single-Run Risk): The primary results in Table 1 (OSWorld) and Table 2 (macOSWorld) report single-point accuracy percentages (e.g., 50.11% vs 44.08%). There is no mention of the number of random seeds used, standard deviations, or confidence intervals. In visual agent benchmarks, pe

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of the experimental results in this paper is insufficient to support the quantitative claims made. The primary issue is the complete absence of uncertainty reporting. Tables 1 and 2 present success rates as exact point estimates (e.g., "59.91%") derived from what appears to be a single experimental run or an unreported aggregation. In the context of

LLM-based agents, where performance is highly stochastic due to sampling, temperature, and non-deterministic environment interactions, a single number provides no information about the reliability of the result. A difference of 6% could easily be noise without a standard deviation or confidence interval.

Furthermore, the paper asserts “consistent improvements” and highlights specific gains without performing any statistical hypothesis testing. There are no p-values, t-statistics, or effect sizes reported to validate whether the observed differences between “MMSkills,” “Text-only,” and “No skill” conditions are statistically significant. The authors also fail to address the multiple comparisons problem; with results reported across 5-6 domains and multiple models, the probability of finding at least one “significant” result by chance is high if no correction (e.g., Bonferroni or Holm) is applied.

Finally, the precision of the reported numbers (e.g., two decimal places for step counts in Table 3) suggests a level of certainty that is not supported by the lack of variance reporting. To make the quantitative claims trustworthy, the authors must re-run experiments with multiple seeds (3), report mean ± SD, and apply appropriate statistical tests with corrections for multiple comparisons.

Action items.

- [**science**] Tables 1 and 2 report single-point success rates (e.g., 59.91%) without any measure of uncertainty (SD, SE, or CI) or number of seeds. In stochastic LLM agent evaluations, a single run is insufficient to distinguish signal from noise. Report mean ± SD over 3 independent seeds for all reported metrics, or explicitly state that results are from a single run and treat them as illustrative rather than definitive.
- [**science**] The paper claims ‘consistent improvements’ and highlights specific gains (e.g., +6.44% in Table 1) but provides no statistical hypothesis tests (e.g., paired t-test, Wilcoxon signed-rank) or p-values to support these claims. Without a test, it is impossible to determine if the observed differences are statistically significant or due to random variance. Add a formal significance test comparing ‘MMSkills’ vs. ‘No skill’ and ‘Text-only’ conditions.
- [**science**] Table 1 and 2 compare MMSkills against baselines across multiple domains (5-6 columns) and models. The paper highlights the ‘best’ performing cells without applying a correction for multiple comparisons (e.g., Bonferroni, Holm, or FDR). With ~12-15 pairwise comparisons, the false positive rate is inflated. Apply a multiple-comparison correction to the reported p-values or rephrase claims to avoid implying significance where none has been tested.
- [**writing**] Table 3 reports ‘Steps’ and ‘ΔSteps’ with two decimal places (e.g., 11.86, -1.25) based on a sample size of 360 tasks. While the mean is precise, the lack of standard deviation or standard error makes the precision misleading. Report the variability (SD or SE) for step counts to allow readers to assess the stability of the efficiency gains.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper demonstrates a high standard of academic writing, with a clear logical flow and well-structured arguments. The introduction effectively sets up the problem of multimodal procedural knowledge, and the methods section provides a detailed, albeit dense, explanation of the framework. The writing is generally precise, and the use of technical terminology is appropriate for the target audience.

However, there are minor issues with sentence structure and flow that, while not impeding understanding, detract from the overall polish. In the Introduction, the transition between specific agent failure modes and the broader claim about human learning is slightly abrupt. In the Methods section, the description of the state card components (Equation 3) suffers from a lack of parallel structure in the subsequent explanatory sentence, making it slightly harder to parse. Additionally, the use of the arrow symbol (->) in the Experiments section to denote percentage changes is informal; standard prose ("from X to Y") is preferred.

The Appendix contains some minor inconsistencies in terminology between table captions and headers (e.g., "median/mean" vs. "Med/Mean"). While these are easily fixable, addressing them would elevate the manuscript's professional quality. The overall readability is strong, and the paper is well-organized, but these small refinements are recommended before final submission.

Action items.

- [**writing**] The paper demonstrates a high standard of academic writing, with a clear logical flow and well-structured arguments. The introduction effectively sets up the problem of multimodal procedural knowledge, and the methods section provides a detailed, albeit dense, explanation of the framework. The writing is generally precise, and the use of technical terminology is appropriate for the target audience. However, there are minor issues with sentence structure and flow that, while not impeding understa

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing with a clear logical flow and well-structured arguments. The introduction effectively establishes the problem of multimodal procedural knowledge, and the methods section provides a detailed, albeit dense, explanation of the framework. The prose is generally precise, and technical terminology is used appropriately for the target audience.

Regarding the prior review, the specific concern about “minor issues with sentence structure and flow” appears to have been addressed or was likely a misinterpretation of the dense technical content. The current text reads smoothly; sentences are grammatically sound, and paragraphs maintain a single focus. Transitions between sections (e.g., from the abstract to the introduction, and from methods to experiments) are logical and well-signposted. The abstract accurately summarizes the method, results, and conclusion without omitting key components.

No new readability issues were introduced in this revision. The text allows a reader to move through the argument without friction, re-reading, or guessing the intent of specific sentences. The writing quality is sufficient for publication.